
PH 611P: Experimental Research Techniques

Course Code	PH 611P
Course Name	Experimental Research Techniques
L-T-P-C	0-0-7-4
Prerequisites	No
Intended for	Core for M.Sc. and I-Ph.D. in Physics
Mutual Exclusion	NA
Approval	10th Senate (Jun 2016)
Revised	64th BoA (May 2026) - syllabus revised

Preamble

According to Newton's third law, one may simply move the earth down and up just by throwing the ball up and down. But the central question is why don't we feel it? It is because of its immeasurability within the uncertainty of the measuring equipment. Performing an experiment without the knowledge of underlying uncertainty has little meaning. The students will be given a flavour of what it really means by (a) performing an experiment (b) developing a mini experimental setup (c) assembling and engineering tools. Aim of the proposed course is to amalgamate the concepts of physics via assembling, developing mini experiments and building components. Students will get to know about a few advanced concepts through high precision instruments.

Course Modules

No

Laboratory Modules

- **Unit I (Mandatory):**
 - **LabVIEW interfacing:** Introduction to data acquisition systems, LabVIEW/open source (such as Python) programming (Interfacing and programming), Signal processing and Error analysis.
 - **Reverse engineering of any vacuum pump:** A rotary pump along with its manual will be provided to the students. They have to dismantle this pump and assemble it and learn about its internal parts. This process helps in the development of skills related with assembling (spatial skills).
 - **Skill development for carpentry:** Self explanatory
- **Unit II:** A student needs to complete at least 3-4 set of experiments from the following list of experiments.
 1. **Temperature dependence of electrical resistivity of materials:** This experiment involves measuring temperature dependent resistivity of any material using four probe method and Vander Pauw methods. The skills that one will develop are to make fine contacts on the sample, learn the intricacies involved in making this set up.

2. **Electronic properties of material using photoemission technique:** Photoemission experiments will be done on any material and its electronic properties will be studied. The skills that one will develop are the intricacies involved in conducting experiments in ultra-high vacuum conditions.
3. **Seebeck coefficient measurement:** Develop mini Seebeck coefficient experiment to distinguish n-type and p-type semiconductors from a mixture of it.
4. **Structural properties of materials using powder x-ray diffraction (XRD) technique:** To understand structural phase transition of any given material. The skills that one will develop are to understand the intricacies involved in any diffraction experiment, temperature variation set up, analysis of any powder XRD pattern.
5. **Experiments on laser physics:** This experiment will be helpful to understand optical pumping, spontaneous emission, induced emission, population inversion, optical resonator, resonator modes etc. The parameters like optimum output coupling, laser threshold and differential efficiency can also be determined. Students will develop skills to handle laser operation by aligning the optical components in the laser cavity.
6. **Two slit interference-one photon at a time:** This experiment will give hands on familiarity to confront wave-particle duality in a precise and definite way. It will develop the skill of single photon detection and theoretical modelling.
7. **Molecular absorption and emission:** The objective is to know the nature of molecular spectra by measuring absorption and emission spectra. This will also provide insights into the molecular energy levels and transitions between them. Preparation of molecular solution and data analysis skills will be developed.
8. **Design and making of any components:** This involves training to make technical drawing, operate lathe for shaping and making any given components.
9. **Scanning tunnelling microscope:** Introduction to the design and working principle and hands on experiments on atomically thin samples.
10. **Magnetic susceptibility of a material as a function of temperature.**

Text Books

1. *Transmission Electron Microscopy*, by C. Barry Carter and David B. Williams, Springer, New York (2016).
2. *Experimental Techniques in Condensed Matter Physics at Low Temperatures*, by Robert C. Richardson and N. Smith; CRC Press (2018).

References

No